

Unit 6, Lesson 8: Digging Deeper into Unit Rates

Benchmarks

MA.6.AR.3.2 Given a real-world context, determine a rate for a ratio of quantities with different units. Calculate and interpret the corresponding unit rate.

MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths and conversions within the same measurement system.

Learning Targets

- ☐ I can determine a unit rate when given different ratios.
- ☐ I can interpret unit rates in real-world contexts.
- ☐ I can use unit rates to solve problems.

6.8.1 Warm-Up: Ticket Booth

At a local fall festival, the following prices are shown at the ticket booth.

Ticket Prices	
1 Ticket	\$0.50
12 Tickets	\$5.00
25 Tickets	\$10.00
50 Tickets	\$25.00
120 Tickets	\$50.00

- Without performing any calculations, explain which option you think would be the best.

- Determine a pair of ratios, tickets:price, that appear to be equivalent. Write them in the space below.

If it is not possible to determine a pair of equivalent ratios from the sign, select one ratio and write another one that is equivalent.

____ ticket(s) : \$_____

____ ticket(s) : \$_____

- By what factor can both values in the first ratio be multiplied to find the second ratio?

6.8.2 Guided Instruction: Interpreting Unit Rate

A **unit rate** is a ratio comparing a number of units of a quantity to ____ unit of another quantity.

1. Several rates are listed. Circle those that represent a unit rate.

earning \$42 for 3 hours

35 miles per gallon

75 miles per hour

2.5 miles in 24 minutes

each banana costs \$0.29

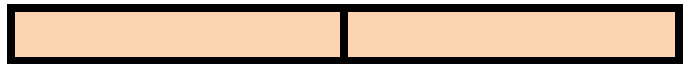
12 inches for every 1 foot

2. Tia was hiking on the Florida National Scenic Trail. She hiked a total of 4.5 miles in 2 hours. The tape diagram models the distance and time she hiked.

Tia hiked 4.5 miles.



Tia hiked for 2 hours.



Notice that each tape diagram is the same size because it represents quantities in the same situation.

- Fill in each box of the tape diagram to show the value it represents.
- What is Tia's unit rate in miles per hour?
- Draw a tape diagram that represents the time it would take Tia to hike 9 miles on the trail at this pace.

3. Carter bikes $12\frac{1}{2}$ miles in 5 hours around the city.
- a. Work with your partner to fill in each box of the tape diagram to show the value it represents. Be sure to include units.



- b. Interpret the unit rate for Carter's bike ride. Include units.
- c. Explain which person is moving slower, Tia or Carter. Include the unit rates in your explanation.

4. Complete the statement.

In each scenario, the unit rate in

☐ hours per mile
☐ miles per hour

 can be determined
by

☐ dividing
☐ multiplying

 the number of

☐ hours
☐ miles

 traveled by the number
of

☐ hours
☐ miles

 traveled.

6.8.3 Guided Practice: Using Unit Rates to Solve Problems

Unit rates would be helpful in considering the ticket booth from the Warm-Up. Miguel set up the table shown to consider the costs per ticket for each advertised price.

1.
- Work with your partner to complete the table. Then, answer the questions that follow.

Total Cost	Number of Tickets	Cost per Ticket
\$0.50	1	\$0.50
\$5.00	12	
\$10.00	25	
\$25.00	50	
\$50.00	120	

- a.
- Using the unit rates, explain which ticket option(s) is(are) the best deal.
- b.
- Using the unit rates, explain which ticket options are the same deal.
- c.
- Compare your answer to the previous question to the equivalent ratios you determined in the Warm-Up. Make any revisions to your previous thinking, as necessary.
- d.
- With your partner, discuss recommendations for how the high school should change its prices for carnival tickets.

2. Holly, Frank, and Ella work part-time for Shoes-R-Us. The table shows how much money each person earns during a work week.

Employee	Weekly Pay	Number of Hours Worked
Holly	\$165	20
Frank	\$162	18
Ella	\$170	20

Work with your partner to answer the following questions using the information in the table.

- Find and interpret the hourly rate Holly earns.
- Which employee earns the higher hourly rate, Ella or Frank?
- If Frank works 25 hours next week, how much should he expect to earn?

6.8.4 Collaboration: Using and Interpreting Unit Rates

Work with your partner to interpret the unit rates for each situation.

1. Tyler paid \$12 for 4 large candy bars.
 - a. Tyler interprets the unit rate as $\frac{\$3}{1 \text{ bar}}$.
His friend Ben interprets the unit rate as $\frac{1 \text{ bar}}{\$3}$. Which person interpreted the unit rate correctly?
 - b. Based on the unit rate, how many candy bars can Tyler purchase for \$45?
2. Aaliyah's family drinks a total of 10 gallons of milk every 6 weeks and 4 gallons of apple juice every 2 weeks.
 - a. Which beverage does Aaliyah's family drink more of in one week?
 - b. How many weeks does it take the family to consume 1 gallon of milk?
3. Carlos buys 24 bottles of spring water for \$3.60 and 3 jars of pasta sauce for \$9.69.
 - a. What does each item, a bottle of spring water and a jar of pasta sauce, cost per unit?
 - b. What would be Carlos' total cost if he purchased 10 bottles of spring water and 2 jars of pasta sauce at their unit prices?

6.8.5 Bring It Together: Interpreting Unit Rates

Work with your partner to write three statements to summarize your learning about unit rates from today's lesson.

1.

2.

3.

6.8.6 Wrap-Up: Ticket Out the Door

A cheesecake recipe says, "Mix 12 ounces (oz) of cream cheese with 15 oz of sugar."

1. How many ounces of cream cheese are needed per ounce of sugar?

2. How many ounces of sugar are used for each ounce of cream cheese?

3. Max plans to make more than one cheesecake. How much cream cheese will she need to mix with 35 oz of sugar?

